Decision Stump

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“Raw” data can often be organized as a table

- Example:

Name	*Phone*	*Weight (kg)*	*Country*
Trump	1 43 22885998	87	USA
Temer	55 23 591927	72	Brazil
Castro	34 95 499402	82	Cuba
…	…	…	…

Each column is a “field”, “parameter”, “feature” or attribute

Each row is a “register”, “example” or “instance”

Attributes or parameters

- Categories:
  - A) Nominal or Discrete, *K* possible values (*K* is an integer)
    - Binary, *K*=2. Example: Brazilian? Yes or No
    - Multiclass, *K*>2. Example: Seven rainbow colors? red, orange, yellow, green, blue, indigo and violet
  - B) Numeric
    - Real numbers. Example: weight 80.7, 23.2, 56.4, ...
    - Integer numbers. Example: age in years 1, 2, 3, ...
  - C) Strings
    - Example: Name John, Paul, Laura, etc.
  - D) Others: date, etc.
- The non-numeric attributes can be mapped to numbers
  - Example: The seven rainbow colors are mapped to integers from 1 to 7
  - Similar to the mapping of a random variable, from an “outcome” (e.g. head or tail) into a number

Classification

Problem:

- Given a vector **x** with *N* parameters, find its class *y* among *Y* possible labels

vector **x** → classifier → *y* or

Fish classifier: (*Y*=2 classes)

or ?

Training set used during the training stage

Length	Weight	Class *y*
12	3.2	
10	0.5	
14	2.8	

Test stage uses "unseen" examples:
x = (13, 4.2)
y = ?

Evaluation: misclassification (or error) rate

Piranha or Pirarucu?
(fishing in the Amazon rivers)

Piranha

Pirarucu

How classifiers are built (trained)?

- There are many automatic classifiers. Examples:
  - If / else rules and thresholds forming a “decision tree”
  - A single if/else rule with a threshold, called “decision stump”
  - K-nearest neighbors (KNN)
  - Artificial neural networks (ANN)
  - Support vector machines (SVM)
  - Naïve Bayes classifiers
  - … and many others
- Each classifier has one or more training algorithms
- Recall: during the supervised training stage, both the parameters **x** and the correct label y is available. During test, the classifier has **x** as input and outputs its guess y_0 for the correct value of y

Simple classifiers (for two simple sets)

- Instead of using an automatic training method, let us design simple classifiers ourselves using the two simple sets below:

Training set

Length	Weight	Class *y*
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

Test set

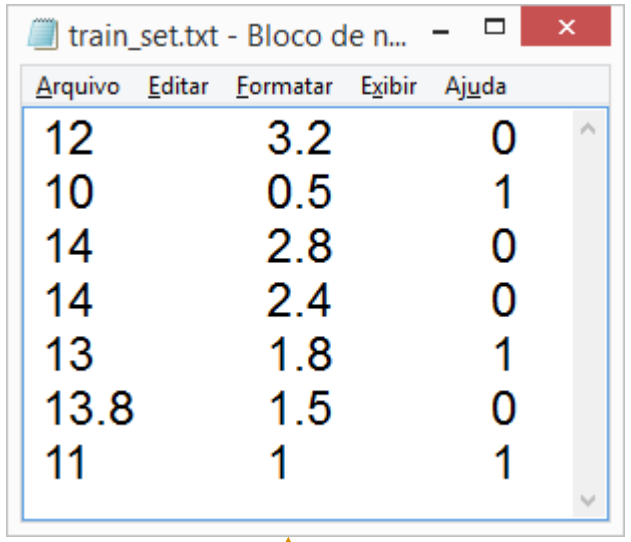
Length	Weight	Class *y*
13	3.1	0
9	0.8	0
12.3	1.4	1
10	2.3	1

Getting familiar with the given data

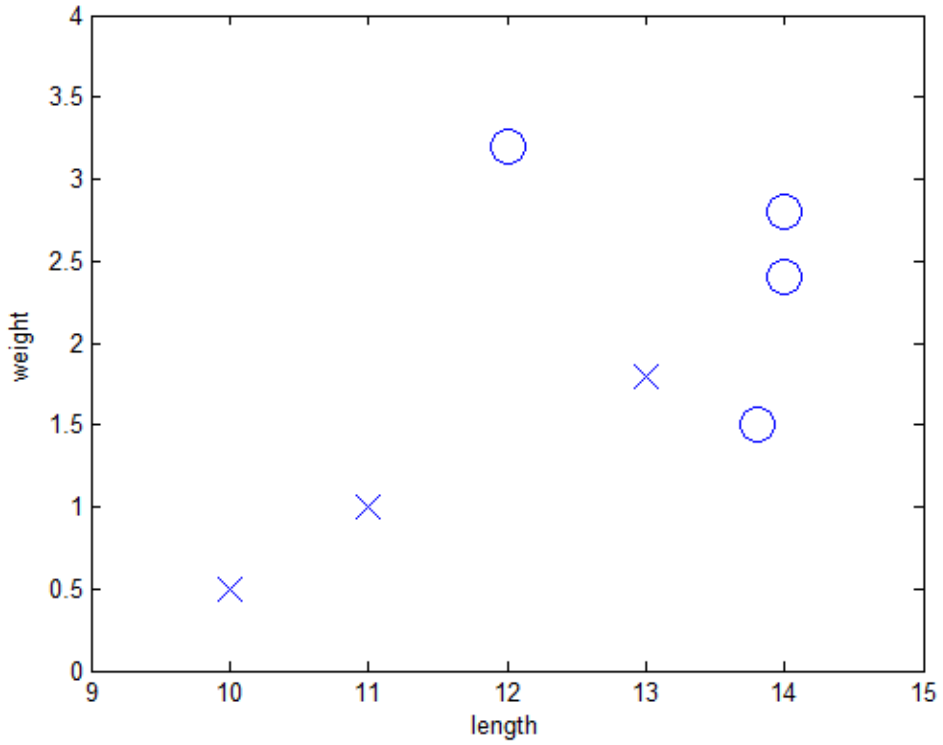
- Because there are (only) two features, it is easy to visualize the training set

Training set

Length	Weight	Class *y*
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1



Text (ASCII) file



Manipulating the data

- Matlab / Octave code showSets.m to plot the data:

train_set.txt - Bloco de n...

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12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

x is a 7x3 array

```
x=load('train_set.txt'); %load from file to memory
%x=load('test_set.txt'); %load from file to memory
indices0=find(x(:,3)==0); %find instances with class label == 0
indices1=find(x(:,3)==1); %find instances with class label == 1
x0=x(indices0,1:2); %label 0: ony two features, without last column
x1=x(indices1,1:2); %label 1: ony two features, without last column
clf %clear figure
plot(x0(:,1),x0(:,2),'o','MarkerSize',14) %plot label 0
hold on %superimpose plots
plot(x1(:,1),x1(:,2),'x','MarkerSize',14) %plot label 1
axis([9 15 0 4]) %resize the x and y axes
xlabel('length'), ylabel('weight')
```

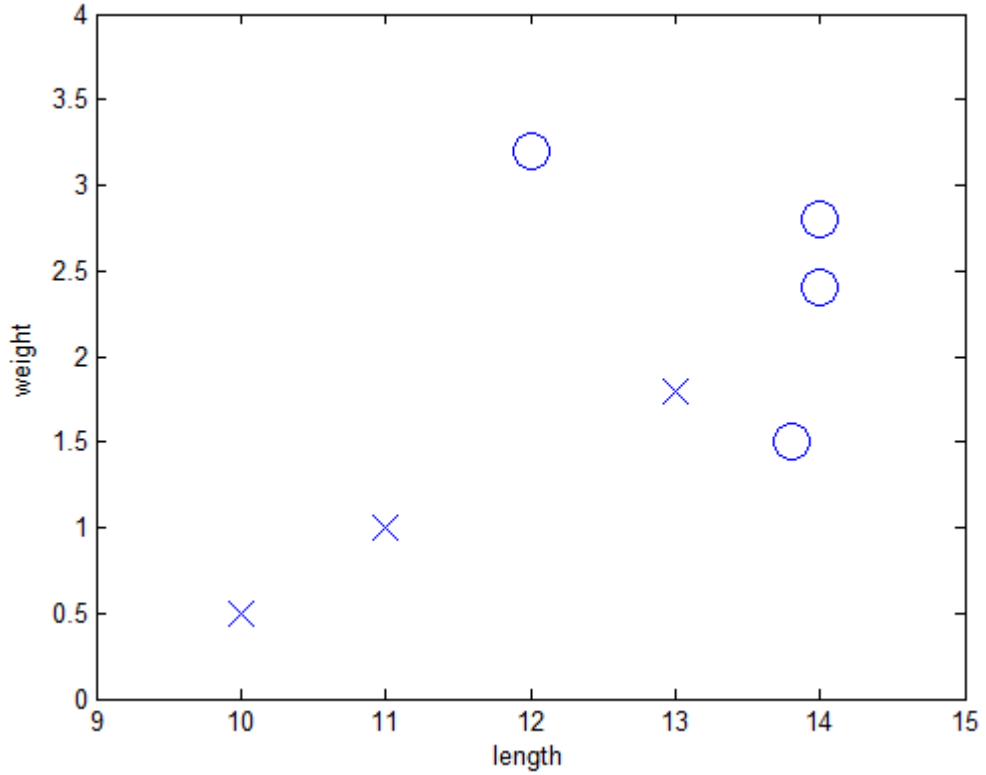
First classifier: decision stu

train_set.txt - Bloco de n...

Arquivo Editar Formatar Exibir Ajuda

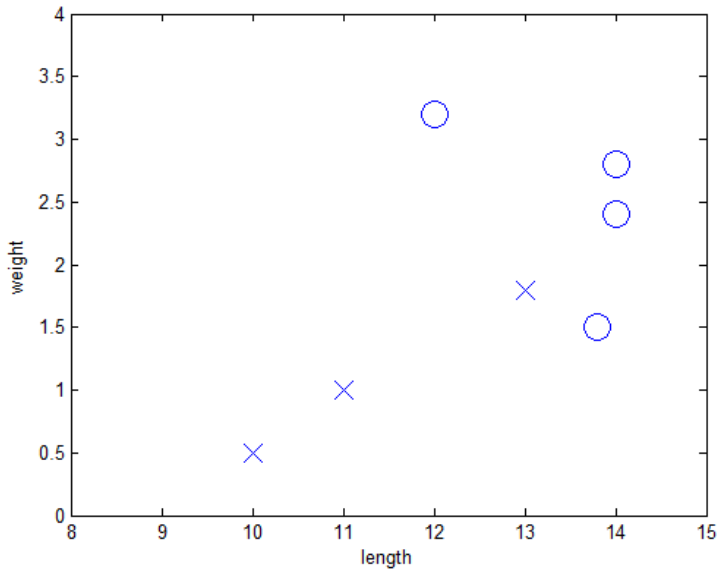
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

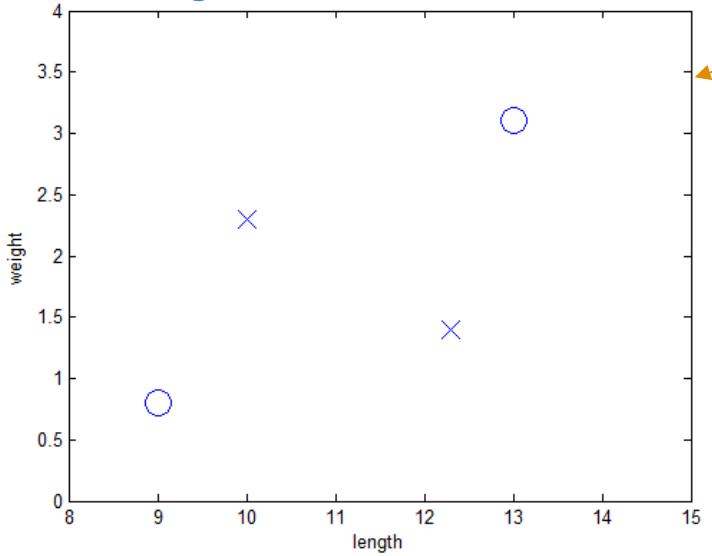
- Decision stump is a single if / else rule based on a chosen threshold value of a chosen feature
- First example:
  - if weight > 1
    - then class label is 0
  - This gives one error in training set!
- Another example:
  - if length < 12
    - then class label is 1
  - This also gives one error in training set!

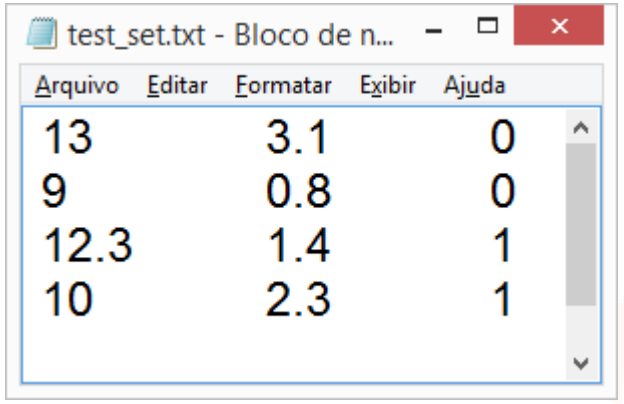


Test our decision stump

- Example:
  - if weight > 1
    - then class label is 0
  - Else
    - then class label is 1
  - This gives one error in training set!
- But gives three errors in this (strange) test set!







Implement (and test): decisionStump.m

showSets.m | decisionStump.m

```
x=load('test_set.txt'); %load from file to memory
[N,M]=size(x); %N is the number of examples and M=3
numberOfErrors=0; %count number of errors
for i=1:N %go over all test set
    exampleFeatures = x(i,1:2);%extract only features (discard label)
    %decision stump: if weight > 1 then class label is 0
    thisWeight = exampleFeatures(2); %chose feature
    if thisWeight > 1
        classifierOutput = 0
    else
        classifierOutput = 1
    end
    if classifierOutput ~= x(i,3)
        numberOfErrors = numberOfErrors + 1; %update number of errors
    end
end
display(['Error rate (%) = ' num2str(100*numberOfErrors/N)])
```